# Checkpoint 3: Averages and the Central Limit Theorem

> **!** Important
>
> **Due 11/29** (12 points). See Project Central for the full project overview. Make sure you have addressed any feedback from Checkpoint 2.

## Goal

Model a **time or measurement** in your scenario with a continuous random variable, then use the Central Limit Theorem to answer a question about an **average** across many people served by your program.

**Course topics used:** continuous random variables, PDFs and CDFs, quantiles, named continuous distributions (Uniform, Exponential, Gamma, Normal), sums of independent random variables, the Central Limit Theorem.

## Your scenario

Use the Checkpoint 3 rows from your Checkpoint 0 table, with any corrections from my feedback.

### Scenario A

#### Community breast cancer screening program

- **Random variable:** $Y$ = size (in mm) of a breast cancer found by screening.
- **Distribution:** Gamma, with the course values mean = 14 mm and SD = 9 mm. For a Gamma, the shape is $(\text{mean}/\text{SD})^2$ and the rate is $\text{mean}/\text{SD}^2$, so use shape $\approx 2.42$ and rate $\approx 0.173$ per mm.

- **Average question:** The program tracks the average size of the tumors it finds each year. Smaller tumors mean the program is catching cancers early. Use $n = 30$ tumors (about one year of cancers found by the program).

### Scenario B

#### County COVID-19 testing and contact tracing

- **Random variable:** $Y$ = incubation period (in days) from infection to the start of symptoms.
- **Distribution:** Gamma, using the mean and SD of the Omicron incubation period from Del Águila-Mejía et al. (2022). For a Gamma, the shape is $(\text{mean}/\text{SD})^2$ and the rate is $\text{mean}/\text{SD}^2$. Calculate the shape and rate from your values, and show your work.
- **Average question:** The department reviews the average incubation period across recent cases to decide how long household contacts should watch for symptoms. Use $n = 50$ cases.
- **Think about:** Your Gamma has a small shape, so it is very right-skewed. Most people get symptoms within a few days, but a few take much longer. Is that realistic for an incubation period? Discuss this in Task 2, and watch how the skew affects your CLT histograms in Task 4.

### Scenario C

#### State newborn heart screening program

- **Random variable:** $Y$ = birth weight (in grams) of a baby born at term.
- **Distribution:** Normal, with the mean and SD for 2005 from Donahue et al. (2010).
- **Average question:** The state reports the average birth weight of babies born in each county every month. Use $n = 100$ babies (a small county's births in one month).
- **Think about:** Your $Y$ is already Normal. In Task 4, explain why your histograms of sample means look Normal even when $n = 5$, and how this is different from a skewed distribution like the Exponential.

## Tasks

### 1. Define your random variable

- Write $Y$ in words, including its units, using the definition for your scenario above.
- State the support of $Y$.

## 2. Justify the distribution

- State the distribution for your scenario and its parameter(s), citing where each value came from.
- Explain why this distribution is a reasonable choice, and where it might not match the real quantity. Consider its shape (symmetric or skewed?), its support, and what we know about the real quantity.
- Plot the PDF of your distribution.

## 3. Solve it analytically

- Find $E(Y)$ and $\text{Var}(Y)$.
- Calculate **two** probabilities or quantiles that answer practical questions for your program, for example:
  - $P(Y > c)$ for a meaningful cutoff $c$ (e.g., a tumor larger than 20 mm, symptoms starting more than 7 days after infection, or a birth weight under 2,500 grams, the cutoff for low birth weight), and
  - the 90th percentile of $Y$.
- Show the setup (integral or CDF) for at least one of them, then compute with R (e.g., `pgamma()`, `pexp()`, `pnorm()`, `qgamma()`, `qexp()`, `qnorm()`).

## 4. Simulate the Central Limit Theorem

Let $\bar{Y}_n$ be the average of $n$ independent observations of $Y$.

- For $n = 5$, 30, and 100, simulate at least 5,000 sample means.
- For each $n$, plot a histogram of the sample means with the normal curve predicted by the CLT, $N\left(\mu, \sigma^2/n\right)$, overlaid.
- Compare the mean and SD of your simulated sample means with the CLT's predictions.
- In 2–3 sentences, describe how quickly the distribution of $\bar{Y}_n$ looks normal. How does the shape (skewness) of $Y$ affect this?

Starter code (Exponential example with a placeholder value; change it for your scenario):

```r
set.seed(550)
rate  <- 1 / 2        # placeholder mean of 2: replace with your distribution
mu    <- 1 / rate
sigma <- 1 / rate
n_rep <- 5000

sim_means <- function(n) replicate(n_rep, mean(rexp(n, rate)))
```

```r
means_30 <- sim_means(30)

hist(means_30, freq = FALSE, breaks = 40,
     main = "Sample means, n = 30", xlab = "Sample mean")
curve(dnorm(x, mean = mu, sd = sigma / sqrt(30)), add = TRUE, lwd = 2)
```

For a Gamma, use `rgamma(n, shape = , rate = )`. For a Normal, use `rnorm(n, mean = , sd = )`.

### 5. Answer a question about an average

Use the CLT to answer a practical question about $\bar{Y}_n$, using the $n$ for your scenario above. For example:

- **Scenario A:** What is the probability that the average size of the next 30 tumors the program finds is more than 15 mm?
- **Scenario B:** What is the probability that the average incubation period across the next 50 cases is more than 3.5 days?
- **Scenario C:** What is the probability that the average birth weight of 100 babies born in one county in a month is less than 3,300 grams?

Then:

- Compute the CLT approximation.
- Check it with your simulation.
- Compare it with the probability that a **single** observation is beyond the same cutoff, and explain why the two answers differ.

### 6. Say it

Write a short paragraph (4–6 sentences) **for the program manager or health department** that explains the difference between what happens to *one* person and what happens *on average* across many people, using your results. Why does this difference matter when the program reports averages to the public?

## Submission

Submit a rendered HTML file (use `code-fold: true`) and your `.qmd`. Make sure the qmd and html are named `lastname_firstinitial_checkpoint_03.qmd` and `lastname_firstinitial_checkpoint_03.html` or `lastname1_lastname2_checkpoint_03.qmd` and `lastname1_lastname2_checkpoint_03.html`. One submission per team. Either person in the pair can submit it.

You can download the `.qmd` editing file for this checkpoint here. The link takes you to GitHub, where you can click the download button (the arrow pointing down) near the top right of the file.

The editing file has the headings and tasks, but not the detailed instructions on this page. **Please do not remove anything from the editing file.** You will only add your work to it.

## Rubric (12 points)

| Criterion | 3 | 2 | 1 |
|---|---|---|---|
| **Setup** (random variable, distribution, justification, PDF plot) | Clear and correct; choice well justified | Minor gaps | Major errors or missing |
| **Analytic solution** (mean, variance, probabilities or quantiles, CLT approximation) | Correct, with work shown | Small errors | Mostly incorrect or no work shown |
| **Simulation** (CLT histograms for three sample sizes, comparison with predictions) | Clear plots; matches CLT predictions; role of skewness explained | Runs, but comparison is weak | Missing or incorrect |
| **Interpretation** (one person vs. the average) | Accurate and in plain language | Somewhat technical or vague | Missing or misleading |